

🛡️ 사이버 보안 Cyber Security 🗝️

✍️ Written by **Donghyun Choi (KGU)**

🌐 - Worldskills Korea - National 2025 (Cyber Security Practices) - 🐼 [Written by NullBins]

- By default, the commands are executed as a root user.

[Project-2] < 🌐 Network security device settings 🐼 >

☆ 추가 수정과제 📖 (Revision Task)

1. UTM SYSLOG 기록 및 전송 (Send UTM SysLog to Server)

- 🌐 UTM(OPNsense 25.1) SYSLOG 전송: UTM SYSLOG 기록을 SERVER에 수집(RSysLog 사용) UTM 발생 로그를 SERVER의 /var/log/opnsense-firewall.log에 저장.

< Configuration >

- [UTM] : OPNsense 25.1
- **System → Settings → Logging**
 - Mode: **Remote**
 - Transport: **UDP(4)**
 - Applications: **All of them**
 - Levels: **All of them**
 - Facilities: **All of them**
 - Hostname: **192.168.1.10**
 - Port: **514**
 - Description: **SERVER**
- [server] : Ubuntu Linux

```
nano /etc/rsyslog.d/50-default.conf
```

```
auth,authpriv.*    /var/log/opnsense-firewall.log
*.*;auth,authpriv.none  -/var/log/opnsense-firewall.log
```

```
nano /etc/rsyslog.conf
```

```
module(load="imudp")
input(type="imudp" port="514")
#$PrivDropToUser syslog
```

```
systemctl restart rsyslog
tail -f /var/log/opnsense-firewall.log
```

- [router]

```
nano /root/icmpLog.sh
```

```
#!/bin/bash
ping -c 1 210.111.10.120
```

```
nano /etc/crontab
```

```
* * * * * root /root/icmpLog.sh; sleep 1
* * * * * root /root/icmpLog.sh; sleep 1
* * * * * root /root/icmpLog.sh; sleep 1
* * * * * root /root/icmpLog.sh; sleep 1
* * * * * root /root/icmpLog.sh; sleep 1
* * * * * root /root/icmpLog.sh; sleep 1
```

< Checking >

- [server]

```
cat /dev/null > /var/log/opnsense-firewall.log
tail -f /var/log/opnsense-firewall.log | grep pass
```

Terminal Image

```

root@server:~# tail -f /var/log/opnsense-firewall.log | grep pass
Oct 5 06:30:37 OPNsense.localdomain filterlog[23858]: 69,,,fae559338f65e11c53669fc3642c93c2,em1,match,pass,out,4,0x0,,64,0,0,DF,17,
udp,56,210.111.10.120,192.36.148.17,4398,53,36
Oct 5 06:30:40 OPNsense.localdomain filterlog[23858]: 69,,,fae559338f65e11c53669fc3642c93c2,em1,match,pass,out,4,0x0,,64,0,0,DF,17,
udp,56,210.111.10.120,192.36.148.17,20980,53,36
Oct 5 06:30:40 OPNsense.localdomain filterlog[23858]: 69,,,fae559338f65e11c53669fc3642c93c2,em1,match,pass,out,4,0x0,,64,0,0,DF,17,
udp,56,210.111.10.120,192.36.148.17,17282,53,36
Oct 5 06:30:40 OPNsense.localdomain filterlog[23858]: 69,,,fae559338f65e11c53669fc3642c93c2,em1,match,pass,out,4,0x0,,64,0,0,DF,17,
udp,56,210.111.10.120,192.203.230.10,35883,53,36
Oct 5 06:30:40 OPNsense.localdomain filterlog[23858]: 69,,,fae559338f65e11c53669fc3642c93c2,em1,match,pass,out,4,0x0,,64,0,0,DF,17,
udp,56,210.111.10.120,192.203.230.10,56173,53,36
Oct 5 06:30:40 OPNsense.localdomain filterlog[23858]: 69,,,fae559338f65e11c53669fc3642c93c2,em1,match,pass,out,4,0x0,,64,0,0,DF,17,
udp,56,210.111.10.120,192.36.148.17,20980,53,36
Oct 5 06:30:40 OPNsense.localdomain filterlog[23858]: 69,,,fae559338f65e11c53669fc3642c93c2,em1,match,pass,out,4,0x0,,64,0,0,DF,17,
udp,56,210.111.10.120,192.36.148.17,17282,53,36
Oct 5 06:30:40 OPNsense.localdomain filterlog[23858]: 69,,,fae559338f65e11c53669fc3642c93c2,em1,match,pass,out,4,0x0,,64,0,0,DF,17,
udp,56,210.111.10.120,192.203.230.10,35883,53,36
Oct 5 06:30:40 OPNsense.localdomain filterlog[23858]: 69,,,fae559338f65e11c53669fc3642c93c2,em0,match,pass,out,4,0x0,,64,1923,0,non
e,17,udp,154,192.168.1.1,192.168.1.10,43119,514,134
Oct 5 06:31:47 OPNsense.localdomain filterlog[23858]: 69,,,fae559338f65e11c53669fc3642c93c2,em0,match,pass,out,4,0x0,,64,1923,0,non
e,17,udp,154,192.168.1.1,192.168.1.10,43119,514,134

```

2-1. UTM 접근 포트 변환 (Convert UTM access port)

- UTM(OPNsense 25.1) WEB 포트 변경: HTTPS 프로토콜 사용. 변경 포트 TCP 50443

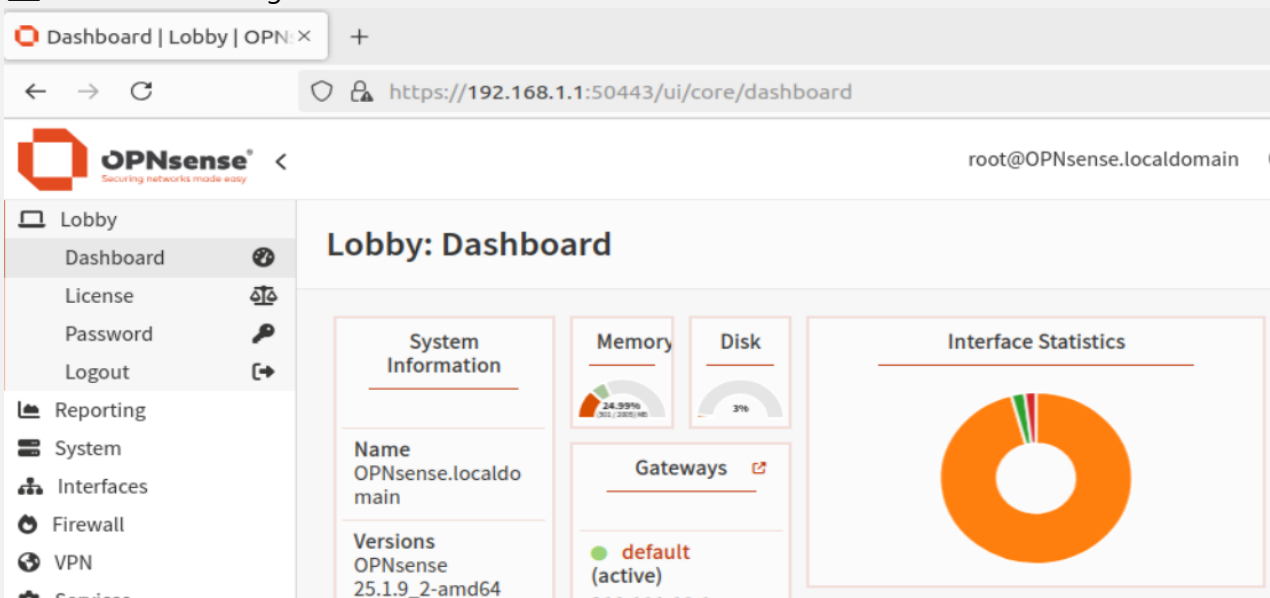
< Configuration >

- [UTM] : OPNsense 25.1
- **System → Settings → Administration**
 - **Web GUI:**
 - Protocol: **HTTPS**
 - Applications: **50443**

< Checking >

- [client]

Web Browser Image



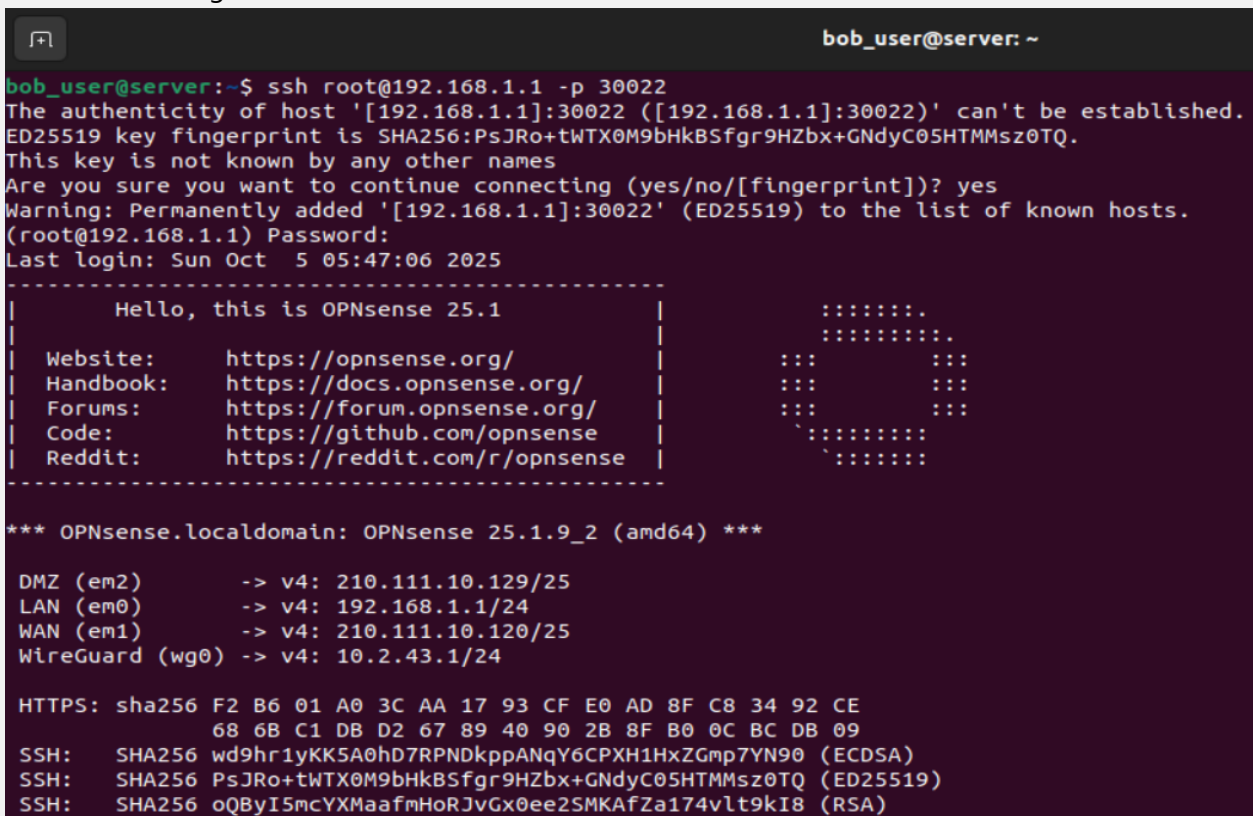
2-2. UTM 접근 포트 변환 (Convert UTM access port)

- UTM(OPNsense 25.1) SSH 포트 변경: SSH 프로토콜 사용. 변경 포트 TCP 30022

< Configuration >

- **System → Settings → Administration**
 - **Secure Shell:**
 - Secure Shell Server: **Enable Secure Shell**
 - Root Login: **Permit root user login**
 - Authentication Method: **Permit password login**
 - SSH Port: **30022**
- [server]

```
ssh root@192.168.1.1 -p 30022
```

 Terminal Image

```
bob_user@server: ~  
bob_user@server:~$ ssh root@192.168.1.1 -p 30022  
The authenticity of host '[192.168.1.1]:30022 ([192.168.1.1]:30022)' can't be established.  
ED25519 key fingerprint is SHA256:PsJRo+tWTX0M9bHkBSfgr9HZbx+GNdyC05HTMMsz0TQ.  
This key is not known by any other names  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '[192.168.1.1]:30022' (ED25519) to the list of known hosts.  
(root@192.168.1.1) Password:  
Last login: Sun Oct  5 05:47:06 2025  
-----  
|           Hello, this is OPNsense 25.1           |  
|-----|  
| Website:   https://opnsense.org/                 |  
| Handbook:  https://docs.opnsense.org/            |  
| Forums:    https://forum.opnsense.org/           |  
| Code:      https://github.com/opnsense           |  
| Reddit:    https://reddit.com/r/opnsense         |  
|-----|  
*** OPNsense.localdomain: OPNsense 25.1.9_2 (amd64) ***  
  
DMZ (em2)      -> v4: 210.111.10.129/25  
LAN (em0)      -> v4: 192.168.1.1/24  
WAN (em1)      -> v4: 210.111.10.120/25  
WireGuard (wg0) -> v4: 10.2.43.1/24  
  
HTTPS: sha256 F2 B6 01 A0 3C AA 17 93 CF E0 AD 8F C8 34 92 CE  
        68 6B C1 DB D2 67 89 40 90 2B 8F B0 0C BC DB 09  
SSH:    SHA256 wd9hr1yKK5A0hD7RPNDkppANqY6CPXH1HxZGmp7YN90 (ECDSA)  
SSH:    SHA256 PsJRo+tWTX0M9bHkBSfgr9HZbx+GNdyC05HTMMsz0TQ (ED25519)  
SSH:    SHA256 oQByI5mcYXMaafmHoRJvGx0ee2SMKAfZa174vlt9kI8 (RSA)
```